

AIMScribe — Guidance Note for CMED

Verifying the clinic and the laptop · where recordings are stored · what the replies mean · AIMS LAB · 5 October 2026

A companion to the **CMED Integration Specification**, answering three questions that came up in planning. Nothing here changes the specification — it explains three things it assumes you already know.

	The question	The short answer
A	How can CMED check which clinic a laptop belongs to, before it starts recording?	Ask the recorder. There is a doctor's command that answers with the clinic code and the doctors registered there
B	Where does every recording end up?	clinic / doctor / date / one file per consultation , named so you can find it without a database
C	The test page shows 200 RECORDING_STARTED and 202 ACCEPTED. What does that mean in a real clinic?	Both green means "stored", not "safe" . A third step you cannot see decides whether the recording is kept

Verifying the clinic and the enrolled laptop

How to make CLINIC_MISMATCH impossible instead of handling it

1 · Why this matters, and what went wrong before

Every AIMScribe laptop is **enrolled to exactly one clinic** when it is installed. That enrolment is the laptop's identity and it cannot be changed from a web page. If CMED sends a `hospital_id` that is not the clinic the laptop belongs to, the recording is **refused outright** with 401 CLINIC_MISMATCH.

This has already happened once in testing. The page was sending HOSP003 after the clinic had been renamed to AALO_DHOLPUR, and every recording was refused until the page was rebuilt. **The lesson is that the clinic code should not be a value CMED stores.** It should be a value CMED asks for.

Who decides what	Why
AIMS LAB enrolls the laptop to a clinic	It happens once, at install, with a single-use token
The laptop knows its own clinic	It is stored on the machine and signed
CMED decides the patient and the doctor	That is the clinical decision, and it is yours
CMED does not decide the clinic	It reads it from the laptop. CMED has no part in enrolment at all

2 · The command that answers it

On the same connection you already use for API 1, send:

```
{ "command": "doctors", "request_id": "who-is-this-pc" }
```

and the recorder replies:

```
{
  "event": "ack",
  "command": "doctors",
  "status": 200,
  "code": "OK",
  "request_id": "who-is-this-pc",
  "data": {
    "hospital_id": "AALO_DHOLPUR",          <-- THIS laptop belongs to THIS clinic
    "doctors": [
      { "doctor_id": "DR0042", "full_name": "Dr Rahman" },
      { "doctor_id": "DR0051", "full_name": "Dr Akter" }
    ]
  }
}
```

Field	What it is	What CMED should do with it
hospital_id	The clinic this laptop is enrolled to	Use it as your hospital_id in all four messages. Do not hardcode a clinic code in the page
doctors	The doctors registered at that clinic — doctor_id and full_name	Use it to fill the doctor selector, so a doctor_id cannot be mistyped

If CMED reads the clinic code from the laptop instead of storing it, CLINIC_MISMATCH becomes impossible — and a clinic can be renamed on our side without CMED changing or rebuilding anything.

3 · The three answers you must handle

What comes back	What it means	What the page should do
hospital_id is a code, doctors has entries	Normal. Enrolled, and we know its doctors	Proceed. Use the code; offer the doctor list
hospital_id is a code, doctors is []	Also normal — a new clinic whose doctors we have not seen yet, or our server briefly unreachable	Proceed. Let the doctor be typed or chosen from your own list. An empty list must never block a clinic
hospital_id is null	This laptop is not enrolled	Do not send API 1. Show "this PC is not registered with AIMS LAB" and contact us

The empty list is the one to get right. It is a list of suggestions, not a list of permissions — CMED decides who is consulting, not us. A shared consulting room that has been offline all afternoon will still answer, from a cached copy, because a stale list of colleagues is more useful than an empty one. But a genuinely new site has nobody in it yet, and the clinic must still be able to work.

4 · The page startup sequence we recommend

	Step	Why
1	Open the connection to ws://127.0.0.1:5050/ws	If this fails, AIMSScribe is not running — say so and stop
2	Send doctors immediately	One round trip on the same machine. It costs nothing
3	Store the returned hospital_id for the session	This is now your clinic code. Never a hardcoded one
4	Fill the doctor selector from the list, if it has entries	Removes mistyped doctor ids, which cause DOCTOR_NOT_AT_CLINIC
5	Show the clinic name on screen , somewhere small	A doctor at the wrong PC notices immediately. This is worth more than it looks
6	Only now allow a patient to be opened	Everything API 1 needs is known

Re-send doctors each time the connection reopens, because a reconnect may mean the recorder restarted, and in principle a laptop could have been re-enrolled between sessions. It is one cheap message and it keeps the page honest.

One thing this does not do
The doctors command tells you the clinic the laptop belongs to. It does not prove the laptop is healthy — that it can reach our server, that its disk has room, that its microphone works. Those show up in the status messages the recorder pushes (section 8 of the specification), and the microphone one is worth displaying: a muted or unplugged microphone records perfect silence and nothing on screen looks wrong.

Where every recording is stored

The pathway from the doctor's laptop to the permanent archive

5 · The three places a recording lives

	Where	What is there	For how long
1	The doctor's laptop the spool folder	The consultation as it is recorded, in 30–90 second pieces	Until the upload succeeds. Holds about 13 hours if our server is unreachable
2	The AIMS LAB server the archive folder	One finished file per consultation , with its clinical record beside it	Until it is copied to cloud storage and verified
3	Cloud storage Cloudflare R2	The permanent copy, as FLAC — same audio, 40% smaller	Permanently. This is the research dataset

A recording is only deleted from one place once the next place has it and has been checked. The laptop will not erase a piece until the server has acknowledged it; the server will not erase a file until the cloud copy has been downloaded again and compared. Nothing is ever deleted on the strength of an upload that was merely accepted.

6 · The folder layout on the server

Four levels, in this order — clinic, then doctor, then the clinic's local date, then one file per consultation:

```
<archive root>/<hospital_id>/<doctor_id>/<YYYY-MM-DD>/<name>/
                                     <name>.wav
                                     <name>.json
                                     <name>.manifest.json
                                     _index.json
```

A real example:

```
/srv/aimscribe/archive/
catalogue.sqlite3          <-- an index of everything below
AAL0_DH0LPUR/
  DR0042/
    2026-10-05/
      P0012345_DR0042_AAL0_DH0LPUR_101432_102755_20261005/
        P0012345_..._20261005.wav          7.6 MB  the audio
        P0012345_..._20261005.json        1.8 KB  the clinical record
        P0012345_..._20261005.manifest.json 6.8 KB  proof it is intact
        _index.json                       0.5 KB  what is in this folder
      P0012387_DR0042_AAL0_DH0LPUR_103010_104402_20261005/
        ...
    DR0051/
      2026-10-05/
        ...
```

6.1 How the file name is built

```
P0012345 _ DR0042 _ AAL0_DH0LPUR _ 101432 _ 102755 _ 20261005
|         |         |         |         |         |
patient  doctor   clinic   started   ended   date
                10:14:32  10:27:55  5 Oct 2026
```

Part	From	Note
patient	your patient_id	exactly as CMED sent it
doctor	your doctor_id	exactly as CMED sent it
clinic	the laptop's enrolment	not the one CMED sent — see Part A
started, ended	the recorder	the clinic's local time , not UTC
date	the clinic's local date	the same day as the folder above it

Why this matters to CMED: you can locate any recording from the five fields alone, without asking us and without a database. If you know the patient, the doctor and the day, you know the folder and you very nearly know the file name.

6.2 The four files in each consultation folder

File	Size	What it is	Why it is there
<name>.wav	~7–120 MB	The audio , one file for the whole consultation	The recording itself, joined from its 30–90 second pieces
<name>.json	~2 KB	The clinical record CMED sent — patient, visit, paramedic, previous_visit, prescription, recording	So the consultation can be read without our database
<name>.manifest.json	~7 KB	Proof the audio is intact — session id, times, timezone, and a signed entry for every piece	Anyone can re-check the audio has not been altered or truncated
_index.json	~0.5 KB	What is in this folder	Lets a folder be understood on its own

Note the extra level: each consultation gets its own folder, named the same as the files inside it. So a day's work is a list of folders, one per patient, not a heap of loose files.

So each consultation is self-contained. One folder, four files, and everything needed to understand *and verify* that consultation is in them. The database is an index into this, not the only copy — if the database were lost entirely, the dataset would still be readable. There is also a `catalogue.sqlite3` at the top of the archive listing every session, clip and pause, which is how a recording is found quickly without walking the folders.

7 · What CMED can and cannot see

	Can CMED see it?	Why
The audio	No. Never	Audio never leaves AIMS LAB's control. CMED is not sent any, and there is no endpoint that would serve it
Whether a recording exists for a visit	Not today	If this would help you, we can add a small endpoint for it. Ask
That a message was stored	Yes	The 202 reply to API 2 and API 3 on Channel B
That recording is in progress	Yes	The status messages on the local connection
The dashboard and the database	No	AIMS LAB only. It holds every clinic's data together

This is deliberate and it protects CMED as much as us. CMED never holds audio of a consultation, so CMED carries none of the obligation that comes with holding it.

What the replies actually mean

Reading the test page, and the third step it does not show you

8 · The two lines on the test page, decoded

recorder: 200 RECORDING_STARTED - Recording.
API 2/3: 202 ACCEPTED - Stored.

The line	Which channel	What it proves	What it does NOT prove
200 RECORDING_STARTED	the browser to the recorder on that PC	The microphone is live and audio is being written to disk	That the consultation will be kept
202 ACCEPTED	your server to the AIMS LAB server	Your message is stored , word for word, and will not be lost	That it matched a recording

Both lines green means both halves arrived. It does not yet mean the consultation is safe. There is a third step, which happens on our server a second or two later and which the test page does not show.

9 · The third step — the match

The recording arrives knowing five things. Your message arrives knowing the same five things. **Our server compares them, and the recording is only kept if they are identical.**

	What happens	Result
1	The recorder starts and tells us: patient, doctor, clinic, start time, date	Recording
2	Your server sends API 2 with the same five fields	Stored
3	We match them	Confirmed — the recording joins the dataset
3*	They do not match, or API 2 never arrives	The recording is kept for 24 hours, then deleted

The failure that matters, stated plainly

If your start_time in API 2 is even slightly different from the one in API 1 — a second out, a missing +06:00, a clinic code from your own list instead of the laptop's — then:

- the recorder says **200 RECORDING_STARTED**
- our server says **202 ACCEPTED**
- **CMED sees nothing wrong at all**
- and the recording is **deleted 24 hours later**

This is the single most important thing in this note. Two green replies do not mean success. They mean both halves arrived; whether they found each other is a separate question that CMED cannot see the answer to.

How to be safe: build the five fields **once**, into one object, at the moment the doctor opens the patient; take the clinic code from the doctor's command rather than from your own settings; and pass that same object to all four messages without recomputing anything. If you do that, the match cannot fail.

10 · What happens when API 2 is late, or never comes

API 2 arrives...	What happens to the recording
within a second or two	Confirmed. Nobody notices anything
late — minutes, or an hour	Recording is never cut. It keeps going, we keep asking, and it is confirmed the moment API 2 arrives
never	Recording is still not cut. It is uploaded and marked <i>unconfirmed</i> , kept out of the dataset, and we are alerted. Deleted after 24 hours

Notice what never happens: a missing message never stops the doctor and never cuts a recording. The clinic keeps working whatever our server is doing. That is why queueing and retrying on your side is the whole of your obligation — a message that arrives an hour late is completely fine, and a message that is never sent is the only real failure.

11 · One real consultation, end to end

Time	Who	What	State
10:14:30	page	connection open; doctors asked and answered	clinic known
10:14:32	page	API 1 to the recorder	recording
10:14:32	your server	API 2 to AIMS LAB	stored
10:14:33	us	the five fields match	confirmed
10:14:35	recorder	first 30–90 second piece uploaded	recording
10:26:55	page	API 3, Channel A — gate armed	still recording
10:26:55	your server	API 3, Channel B — the prescription	stored
10:27:10	doctor	prints it, hands it over, counsels the patient	still recording — the valuable part
10:27:55	page	API 1 for the next patient	this one closes , next one opens
10:28:10	server	pieces joined into one file, named, checked	archived
10:30:00	server	copied to cloud storage and verified	permanent

The consultation lasted 13 minutes and 23 seconds. **CMED sent four messages, none of which the doctor waited for.**

12 · The difference between the test page and a real clinic

	On the test page	In a real clinic
The clinic code	typed into the page, or built into it	read from the laptop with the doctors command
The recorder	may be the only one running	fourteen rooms, each its own laptop and its own enrolment
The match	not shown	decides whether the recording lives. Section 9
API 2 timing	sent by hand, in order	sent by your server, possibly late, possibly retried
A failure	you see it on screen	often silent. Which is why the morning check on unconfirmed recordings exists
The microphone	you are watching it	nobody is. Show the level meter

13 · One page to remember

Question	Answer
How do I know which clinic a laptop is?	Send {"command": "doctors"} and read hospital_id
Should I store the clinic code in my page?	No. Ask the laptop every time the connection opens
What if doctors comes back empty?	Normal. Proceed. Never block the clinic
What if hospital_id is null?	Not enrolled. Do not record; contact AIMS LAB
Where is a recording stored?	clinic / doctor / date / patient_doctor_clinic_start_end_date.wav
What is beside it?	A .json of the same name, with the clinical record CMED sent
Does CMED ever get audio?	No. Never
Does 200 RECORDING_STARTED mean safe?	No. It means the microphone is live
Does 202 ACCEPTED mean safe?	No. It means stored, not matched
So what makes it safe?	The five fields matching between API 1 and API 2
What if they do not match?	Silently deleted after 24 hours. CMED sees two green replies
How do I make that impossible?	Build the five fields once and reuse the same object for all four messages
What if API 2 is an hour late?	Completely fine. It is confirmed when it arrives
What if it is never sent?	The recording is lost. Queue and retry

Prepared by AIMS LAB, United International University. Companion to the CMED Integration Specification. Every behaviour described here is taken from the code that implements it.